

Aleksandrs Drozdovs

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PROFESSIONAL SUMMARY

Computer Science & Software Engineering graduate (Maynooth University, September 2026, 2:1 Honours), available now for graduate software engineering and quantitative developer roles. Built a six-phase, pre-registered research program (176-test CI) that honestly disproved a \$28,400 backtest edge across ~10 years and three futures markets; earned ~\$15,000 in live trading profit at DLT Capital; five fixes merged upstream into established open-source libraries. Ships full-stack TypeScript/Next.js and Python products and Go systems code with tests and CI.

WORK EXPERIENCE

DLT Capital

Feb 2025 - Jul 2025

Quantitative Researcher & Software Engineer (Part-Time)

Maynooth, Ireland

- Researched and implemented systematic crypto and futures trading strategies that generated ~\$15,000 in live trading profit, applying statistical filters and strict risk rules; built the backtesting software used to validate them.

PROJECTS

Open Source Contributions - 5 merged upstream PRs

Merged under maintainer review: a schema-ordering bug in narwhals' pandas-like backend (six new tests, each proven failing first), a Dubai Financial Market trading calendar built from scratch for exchange_calendars (154-test suite), a failure-classification fix in a robotics SDK, and docs/localisation fixes in yfinance and holidays.

hflow #288, narwhals #3856, exchange_calendars #593, holidays #3748, yfinance #2936

jobq - Durable Job Queue in Go

Durable job queue from scratch: at-least-once delivery, actor-per-topic concurrency, leases, dead-letter queues and a CRC-checked, group-committed write-ahead log (550µs to 65µs per op under 16 producers). Crash harness across 5 rounds: zero jobs lost or resurrected.

Go (standard library only), write-ahead log, property-based testing - github.com/aleks-drozy/jobq

Options Pricing Engine

Three independent option pricers cross-validated through seven machine-checked numerical gates (put-call parity to 2.84e-14). Inverted a real SPY option chain to implied volatility, recovering the smile flat-vol Black-Scholes cannot explain; the live explorer re-verifies every gate on load.

Python, Black-Scholes, CRR binomial, Monte Carlo - github.com/aleks-drozy/options-pricing-engine

Quantitative Strategy Research Program

Six-phase, pre-registered program (176-test CI): no-lookahead backtest engine, leak-free walk-forward optimisation and cluster-bootstrap confidence intervals proved a \$28,400 backtest edge was not real across ~10 years and three futures markets (1,402 out-of-sample trades); the runner hash-verifies a frozen config before it will run.

Python, pandas, NumPy, pytest, GitHub Actions - github.com/aleks-drozy/fyp-strategy-engine

EDUCATION

B.Sc. (Hons) Computer Science and Software Engineering [Maynooth University](#)

Sept 2026

2:1 (Upper Second-Class Honours). Modules: Machine Learning & Neural Networks, Parallel & Distributed Systems, Algorithms & Data Structures, Software Testing & Verification.

SKILLS

Languages: Python, TypeScript, JavaScript, Go, Java, C++, SQL

Frameworks & Tools: React, Next.js, Node.js, FastAPI, pandas, NumPy, scikit-learn, pytest, Vitest, Playwright, PostgreSQL, Supabase (RLS), DuckDB, dbt, GitHub Actions, Vercel

AI/ML & Stats: LLM integration (Anthropic, Groq), AI agents (MCP, Claude Code; Anthropic Academy certified), applied ML, bootstrap/Monte Carlo statistics

Interests: Judo black belt (16 yrs, multiple Irish national medals); funded algorithmic-trading account with verified payouts